🏆 FINTECHATHON 2026 · TOPIC E · INTERNATIONAL TRACK

CenterPoint ProERP

Embedded AI SME Financial Copilot

Bridging Real-Time Enterprise General Ledgers with WeBank Digital Financial Inclusion

中文副标题：贯通企业级真实复式记账总账与微众银行小微数字普惠金融

100%
TEST PASS RATE (15/15)
PHPUnit Feature & Unit green

< 50ms
LEDGER AGGREGATION
Zero-hallucination DB speed

3 Tiers
SUPERVISED SECURITY
Human-in-the-loop auth modal

4 Products
WEBANK SME MATCHING
Factoring, WC loan, Stock credit

Host: Tsinghua University PBCSF | Strategic Partner: WeBank | Target: Manufacturing, Retail & Supply Chain SMEs

THE SME REALITY

The Liquidity Blindspot & The Bank Credit Gap

82% of SME failures stem from cash flow mismanagement, not lack of demand.

⚠️ The SME Internal Dilemma

- **Operational Blindness:** Vital runway and burn metrics remain trapped in manual bookkeeping spreadsheets.
- **Receivables Drag:** 30–60 day payment delays from enterprise buyers lock up working capital.
- **Desperation Timing:** SMEs only approach lenders after entering insolvency, when it is too late.
- **Payroll Cliff:** Wage commitments occur on fixed dates regardless of customer collection cycles.

🏦 The Commercial Bank Friction

- **High Underwriting Costs:** Manual due diligence exceeds $2,500/loan, making sub-$50k tickets unprofitable.
- **Collateral Obstacle:** Legacy underwriters require physical real estate mortgages that asset-light SMEs lack.
- **Lack of Audited Statements:** 90% of micro/small enterprises cannot produce audited multi-year accounts.
- **Post-Disbursement Blindness:** Banks have zero visibility into daily solvency after capital disbursement.

CORE INNOVATION

Embedded ERP Intelligence vs Generic Chatbots

Why fintech requires deterministic ledger grounding instead of decorative chatbot wrappers.

ARCHITECTURE DIMENSION	GENERIC LLM CHATBOTS (CHATPDF / WRAPPERS)	CENTERPOINT PROERP COPILOT (TOPIC E)
Data Foundation	Stale, unverified text PDFs & approximations	Live Double-Entry Accounting Ledger
Hallucination Risk	High; hallucinates phantom balances & rates	Zero; strict Function Calling DB aggregates
User Presentation	Unstructured markdown text paragraphs	6 Interactive High-Fidelity ERP Cards
Operational Scope	Passive reading; cannot perform operational actions	Supervised Tier 2 Human-in-the-Loop Actions
Tenant Isolation	Vulnerable vector search; high leakage risk	Cryptographic JWT Scope (BelongsToTenant)

ENGINEERING BLUEPRINT

Domain-Driven Modular Monolith Architecture

Dual-engine coordination combining live GPT-4o API with deterministic offline sandbox.

1. Presentation Layer

React 19 + TypeScript

- Zero-overflow layout (total width ≤ 1030px)
- Symmetrical action buttons (Inspect + Actions ▾)
- Streamlined CopilotChat & WeBank Matcher tabs

2. Orchestration Core

Laravel 13.x Modular Backend

- AgentOrchestrator with anti-injection filter
- AgentPermissionGuard (Tiers 0/1/2/3)
- Dual engine: OpenAI GPT-4o & sandbox fallback

3. Data & Isolation

PostgreSQL / MySQL / SQLite

- Strict `BelongsToTenant` scope on all models
- Verified cross-tenant 404 rejection (Test T02)
- Immutable `agent_audit_logs` compliance trail

CORE ENGINE I

Real-Time Cash Runway & Float Velocity

Continuous calculation of liquid cash reserves against historical operating burn velocity.

CashPositionCard Live Output

BDT 250,000
TOTAL LIQUID CASH

16 Days
OPERATING RUNWAY

BRAC Bank Principal	BDT 145,000.00
City Bank SME Payroll	BDT 85,000.00
Petty Cash Float	BDT 20,000.00

Algorithmic Formulation

$$\text{Daily Burn} = \frac{\sum \text{Operating Expenses (90d)} + \text{Payroll}}{90} = \text{BDT } 15{,}200 / \text{day}$$ $$\text{Runway Days} = \left\lfloor \frac{\text{Liquid Cash}}{\text{Daily Burn}} \right\rfloor = \mathbf{16 \text{ Days}}$$

⚠ **Moderate Caution Zone:**
Runway is below the 30-day safety buffer with month-end factory payroll obligations due in 14 days.

CORE ENGINE II

30-Day Deficit Stress Testing & Debt-Aging Analysis

Forward-looking cash flow forecast with probability-weighted debtor recovery curves.

Probability-Weighted Aging Matrix

AGING BUCKET	OUTSTANDING	FACTOR	EXPECTED RECOVERY
1 - 30 Days	BDT 150,000	95%	BDT 142,500
31 - 60 Days	BDT 100,000	80%	BDT 80,000
61 - 90 Days	BDT 120,000	50%	BDT 60,000
90+ Days	BDT 60,000	20%	BDT 12,000

Projected Liquidity Deficit Alert

- BDT 145,000

PROJECTED DEFICIT AT DAY 14

Stalled debtor concentration (BDT 430,000 across Apex Superstore & Agora Retail) delays cash inflows past the month-end worker wage disbursement date.

Critical Action Required | Recommended: Invoice Factoring

CORE ENGINE III

WeBank SME Digital Financing Match Engine

Algorithmic pre-qualification across 4 tailored digital lending products based on ledger health.

95% Match #1

Digital Invoice Factoring

BDT 344k

MAX LIMIT (80% AR)

- Interest: 6.8% APR
- Turnaround: Instant Same-Day
- Collateral: Verified AR Invoices

91% Match #2

Working Capital Term Loan

BDT 1.5M

MAX LIMIT (3X MO REV)

- Interest: 7.5% APR
- Turnaround: 24 Hours
- Collateral: Cash Flow Verification

84% Match #3

Inventory Asset Credit Line

BDT 600k

MAX LIMIT (60% LTV)

- Interest: 8.2% APR
- Turnaround: 48 Hours
- Collateral: Warehouse Stock Pledge

78% Match #4

Equipment Leasing Line

BDT 2.4M

MAX LIMIT (24X PROFIT)

- Interest: 9.0% APR
- Turnaround: 3–5 Days
- Collateral: Machinery Hypothecation

SECURITY & SAFETY

Three-Tier Supervised Permission Architecture

Agent PROPOSES. Human CONFIRMS. Probabilistic models never commit unilateral mutations.

Tier 0: Read-Only

Autonomous Retrieval

`get_cash_position`, `list_overdue_invoices`, `get_pl_summary`

Zero database writes. Safe for automatic execution on chat invocation.

Tier 1: Advisory

Non-Binding Advice

`match_financing_options`

Ranks WeBank SME products. Writes informational assessment record with zero financial liability.

Tier 2: Supervised

Human Confirmation

`create_cash_flow_alert`

Agent execution is halted. Modal renders exact parameters. Requires named human supervisor signature.

⛔ **Tier 3 (Autonomous Commitment): PERMANENTLY BLOCKED**
Automatic loan contracting, fund disbursements, or banking transfers are strictly prohibited at the kernel level.

COMPLIANCE & GOVERNANCE

Multi-Tenant Cryptographic Isolation & Auditability

Zero cross-tenant leakage with tamper-evident execution tracing.

Tenant Isolation Invariants

- **JWT Claims Only:** `tenant_id` is extracted strictly from cryptographically signed tokens.
- **Global Query Scope:** All Eloquent models inherit `BelongsToTenant` (`WHERE tenant_id = ?`).
- **Integration Test T02:** Verifies cross-tenant data requests trigger immediate 404 rejection.

Canonical ERP Audit Trail

TIMESTAMP	ACTION	TIER	LATENCY
17:15:22	get_cash_position	Tier 0	42 ms
17:16:04	match_financing	Tier 1	88 ms
17:18:10	create_alert	Tier 2	114 ms

CONCLUSION & VISION

Production Quality & WeBank Ecosystem Partnership

Ready for deployment across global emerging-market SMEs.

Verified Quality

- **15/15 Backend Tests:** 100% PHPUnit pass rate (58 assertions)
- **7/7 Frontend Tests:** Vitest suite clean in 1.8s
- **0 TypeScript Errors:** Strict typecheck passed

Performance Metrics

- **42 ms:** Local ledger database aggregation
- **172 ms:** Full demo seeder execution
- **1.2 s:** Cloud GPT-4o end-to-end response

WeBank Roadmap

- **Q1 2027:** Direct WeBank Open API Webhook integration
- **Q2 2027:** Automated digital pre-disbursement
- **Q3 2027:** Cross-border supply chain factoring

Empowering SMEs with Real-Time Financial Resilience

Thank you, Tsinghua University PBCSF & WeBank Organizers.